Pxl 3/24/5/6

Index Number:_____ Programme:_____

UNIVERSITY OF ENERGY AND NATURAL RESOURCES, SUNYANI

SCHOOL OF ENGINEERING

DEPARTMENT OF COMPUTER AND ELECTRICAL ENGINEERING

LEVEL 200 END OF FIRST SEMESTER EXAMINATION, 2020/2021

Bachelor of Science (Electrical and Electronics Engineering)

Bachelor of Science (Computer Engineering)

CENG 205: Computer Programming For Engineers

April, 2021 **Time: 2 Hrs : 40 Mins**

Materials required: *Pen, Pencil, Eraser and Ruler. The use of calculator is NOT allowed in this paper.*

Instructions: *This paper consists of TWO SECTIONS (A and B). Answer all questions in SECTION A and B.*

SECTION A – Answer all questions. *Shade completely the letter corresponding to the correct option from A to D on the Shading Answer Sheet provided.* **Any correct answer carries 1 mark.**

1. All the following are C++ data types except ____.
   A. char
   B. bool
   C. int
   D. main

2. The sequence of logically related instructions that allow a computer to perform specific task is called _____.
   A. program
   B. process
   C. source code
   D. language

3. _____is a structured data type of similarly typed elements which are accessible using indices.
   A. Unio
   B. Structures (Records)
   C. Set
   D. Array

4. String characters are always terminated by appending the _____.
   A. \0 - character
   B. & - character
   C. ; - character
   D. \n - character

5. What is the value printed for variable `delta` if the following code is executed?

```
int delta = 0, x = 3;
if(x % 2 == 1)
     delta
x++;
if(x
     delta
x++;
if(x
     delta
cout          ;
```

   A. 0
   B. 7
   C. 6
   D. 4

6. Indicate the false statement.
   A. Value in `&a[0]` is `a[0]`
   B. `&a[3]` is equivalent to `(a + 3)`
   C. `a[3]` is equivalent to `*(a + 3)`
   D. Value in `&a[0]` is `*a`

7. Using `enum sports {BASKETBALL, FOOTBALL, HOCKEY, BASEBALL, SOCCER, VOLLEYBALL};` Assume that `popularSport` is declared base on the enum **sports** and intialize to **FOOTBALL**. What is the value of `popularSport` if the code below is executed?

```
popularSport
<sports>(popularSport
```

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A. BASKETBALL

B. HOCKEY

C. 4

D. SOCCER

8. In a class definition, member functions designed to modify the data members of the class are called accessor functions. Indicate whether statement true or false.

A. True B. False

9. What would be printed by the following statements?

```
double *pt;
double a[3] = {1.2, 2.3, 3.4};
pt = &a[1];
pt+=1;
cout<<*pt<<endl;
```

A. 3.4

B. 2.2

C. 4.4

D. 1.2

10. The selection logical construct is implemented in C++ using a/an _____.

A. assignment statement

B. if statement

C. while statement

D. jump statement

11. The actual argument in a function call have to match the _____ of the parameters that appear in the function definition.

A. order

B. data type

C. number

D. both (A) and (B)

12. The modulus operator " % " is used in division to return the

A. quotient

B. dividend

C. reminder

D. divisor

Programme:_____

13. A procedural set of instruction or statements, that may take argument then perform some work, and may return a result.

A. Function

B. Property

C. Object

D. Class

14. Consider the program segment given below. Its output can best be described as:

```
char a='b'; char b='c';
char c=a;
cout<<'a'<<b<<'c'<<a<<'b'<<c;
```

A. aacbbc

B. accbbb

C. accabb

D. abcabc

15. In the following code fragment, how many times does sum get incremented?

```
for(int i=0;i<3;i++)
{
  for(int j=0;j<3;j++)
  {
    sum++; // line X
  }
}
```

A. 9

B. 2

C. 3

D. 4

16. _____ is a kind of placeholder that is filled in a function argument when the function is called.

A. Actual parameter

B. Formal parameter

C. Declaration

D. Literal

17. For a list of length *n*, selection sort makes exactly _____ key comparisons and *3(n -1)* item assignments.

A. $n(n + 1) / 2$

B. $3(n - 1) / n$

C. $n(n - 1) / 2$

D. $(n + 1) / 3$

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18. In array declaration, the name of the array points to the address of the entire array. Indicate whether statement true or false.

A. True B. False

19. Suppose the following codes were executed:

```
int a, b, c;
a = 1; b = 4;
while(a < b) {
        a = 2 * a;
        b = b + 1;
}
z = b;
```

What is the value of z ?

A. 4
B. 8
C. 7
D. 5

20. What is wrong with the following piece of code fragment?

```
string name;
name = "John"; // line 2
switch(name)
{
  case "Smith": cout << "smith";
  break;
  case "John": cout <<"john";
  break;
}
```

A. Strings cannot be used with switch statement
B. Default statement is missing
C. There is an error in line 2
D. Semi colon is missing at the end of the switch statement

21. Sorting is a fundamental problem associated with arrays that seeks to _____.

A. insert into the array the target element
B. locate in the array where target element is
C. reorder the elements in the array
D. all the above

22. What is the output, if the following code fragment is executed?

```
...
string str="John Smith";
cout << str.length();
...
```

Program

A. 12
B. 13
C. 10
D. 9

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23. The output of the program segment below is the text: **greater than Done.** Indicate whether statement true or false.

```
int x = 3;
if ( x <= 3)
     if ( x < 3)
          cout << "less than ";
else
     cout<<"greater than ";
cout<<"Done.";
```

A. True B. False

24. What would be printed by the following program?

```
...
int add1(int &n) {
     n++;
     return n; }
int main() {
     int m = 0;
     for(int j = 0; j < 5; j++)
          m += sub1(j);
     cout << m << "\n";
     return 0; }
```

A. 0
B. 1
C. 9
D. Compile error

25. Which of the following is true?

A. Actual parameters are present in the function declaration and formal parameters are present in the function definition.
B. Actual and formal parameters are both present in the call to the function.
C. Actual parameters are present in the call to the function and Formal parameters are present in the definition of the function.
D. Formal parameters are present in the call to the function and the actual parameters are present in the definition of the function.

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26. How many times is the body of the loop executed?

```
int i=1;
while(true)
{
    cout << i;
    if(i==5)
    break;
    i++; }
```

A. 5 times

B. 10 times

C. Forever

D. 0 times

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27. Suppose the code snippet is executed. What would be the value of z?

```
int x, y, z;
x =3; y = 7; z = 9;
if (x > y)
    if (x - y == 4)
        z = 33;
    else
        z  = 77;
```

A. 4

B. 9

C. 77

D. 33

28. What would be printed by the following program?

```
...
double triangle(double base,
double height = 10.0);
int main()
{
  cout.setf(ios::showpoint);
  cout<<triangle(10.0)<<"\n";
  return 0;
}
double Triangle(double base,
double height)
{
  return 0.5*base*height;
}
```

A. 50.0

B. 50

C. 50.0000

D. Compile error

29. Which of the following sequence is CORRECT in C++ compilation process?

Programme:_____

A. Object code → Compiler → Linker → Source code → Executable program

B. Object code → Source code → Compiler → Linker → Executable program

C. Source code → Object code→ Linker → Compiler → Executable program

D. Source code → Compiler → Object code → Linker → Executable program

30. Which of the following is true?

A. Actual parameters are present in the function declaration and formal parameters are present in the function definition

B. Formal parameters are present in the call to the function and the actual parameters are present in the definition of the function

C. Actual parameters are present in the call to the function and Formal parameters are present in the definition of the function

D. Actual and formal parameters are both present in the call to the function

31. If I wanted to execute some piece of code based on what range an *angle* variable is, which of the following statements would I use?

A. while statement

B. if statement

C. switch statement

D. for statement

32. What is the only function that all C++ programs must contain?

A. function ()

B. program ()

C. start ()

D. main ()

33. The output of the program below is **10 50**. What is the input values of a and b?

```
...
// Function definition
void f(int &a, int b, int &c) {
 b = a - c; a = b * c;
}
// Content of Main function
...
 int a, b;
 cout<<"Enter a and b";
 cin>>a>>b;
```

```
f(b,a,a);
cout << a << ' ' << b << endl;
...
```

A. a = 25; b = 10

B. a = 10; b = 15

C. a = 15; b = 10

D. Compile error

34. What does the expression `! ( true && ! (false || true ))` evaluate to?

A. True B. False

35. What is wrong with the following piece of code fragment?

```
string name;
name = "John"; // line 2
switch(name)
{
  case "Smith": cout << "smith";
  break;
  case "John": cout <<"john";
  break; }
```

A. Semi colon is missing at the end of the switch statement

B. Default statement is missing

C. There is an error in line 2

D. Strings cannot be used with switch statement

36. What is the correct value to return to the operating system upon successful completion of a program?

A. Programs do not return a value

B. -1

C. 0

D. 1

37. What is wrong with the following code fragment?

```
int main() {
  int x=4,y=3;          // line 1
  const int i = 10;     // line 2
  x = x + i;            // line 3
  i = y*2; }            // line 4
```

A. There is an error in line 3

B. There cannot be 3 variables in such a simple program

C. There is an error in line 4

D. Variables are not declared before use

38. For the following code fragment, how many times does sum get incremented? In other words how many times is line X executed in the following program?

```
for(int i=0;i<3;i++)
{
   for(int j=0;j<3;j++)
   {
     sum++; // line X
   }
}
```

A. 2

B. 4

C. 9

D. 3

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Consider execution of the following switch statement:

```
int Enter = 10;
cin >> Enter;
switch (Enter)
{
     case 1: Enter = -4;
     case 2: Enter = -6;
     case 4: break;
     case 6: Enter = -8;
     break;
     default: Enter = -1;
}
```

Answer the following questions:

39. What would the value of Enter be after execution of this code if the value read for Enter were 4?

A. None of the above

B. 10

C. -6

D. -4

40. What would the value of *Enter* be after execution of this code if the value read for *Enter* were 1?

A. -6

B. -4

C. None of the above

D. -8

41. Which of the following is complete function?

A. void funct (int) { cout<<"Hello" }

B. int funct ( );

C. int funct (int x) { return x++; }

D. void funct (x) { cout << "Hello"; }

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42. Suppose we need to sort a list of employee records in ascending order, using their social security number (a 9-digit number) as the key (i.e., sort the records by social security number). If we need to guarantee that the running time will be no worse than *n log n*, which sorting methods could we use?
    A. Merge sort
    B. Quick sort
    C. Insertion sort
    D. None of the above

43. What is required to avoid falling through from one case to the next?
    A. Exit ();
    B. you need a semicolon
    C. end;
    D. break;

44. What would be printed by the following statements?

```
int x = 1;
if (x = 2)
    cout << "x = 2.\n";
else
    cout << "x is not 2.\n";
```

    A. x is not 2.
    B. x = 2.
    C. compile error
    D. None of the above

45. This is a reason why C++ is still a viable programming language some twenty years after its creation:
    A. All of these are good reasons.
    B. There is a substantial legacy library of useful C++ code
    C. There are lots of smart compilers for it that produce efficient code.
    D. It is a concise language.

46. What is the logical condition under which the following while loop will terminate?

```
int Beta = 5;
while (Beta >= 0 && Beta < 10)
{
cout << Beta << endl;
cin >> Beta;
}
```

    A. Beta <0 ||Beta >= 10
    B. Beta <= 0 && Beta > 10
    C. Beta <0 &&Beta >= 10
    D. All of the above

47. What do the following two *cout* statemen out?

```
{
  int i = 1;
  double x = 1.111;
  cout <<i<<" "<<x<<"\n";
  {
   int x = 2;
   double i = 2.222;
   cout <<i<<" "<<x<<"\n";
  }
}
```

    A. 1 1.111
       2.222 2
    B. 1 1.111
       2 2
    C. 1.111 1
       2.222 2
    D. 1 1
       2.222 2

48. What is the return type of the function with prototype: `int func (char x, doub v, float t);`
    A. double
    B. int
    C. float
    D. char

49. Which of the following is an illegal operation declaration in C++?
    A. `float arr[2] = {2.5, 5};`
    B. `mySport = popularSport + 2;` //Assume mySport and popularSport are enum declaration on *sports*
    C. `char x;`
       `switch (x) { case const:`
       `statement1; break; }`
    D. None of the above

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50. Which of the following is a correct comment?
    A. /* Comments */
    B. {/ Commnets }
    C. */ Comments */
    D. ** Comments **

51. Which of the following is a properly defined structure?
    A. struct a-struct { int a; };
    B. struct a_struct int a;
    C. struct a_struct { int a };
    D. struct { int a; }

52. How does the computer store the letter 's'?
    A. In ASCII representation
    B. In binary format
    C. The letter is mapped onto some number and this number is stored
    D. All of the above

53. The actual arguments in a function call have to match the _____ of parameters that appear in the function definition.
    A. data type
    B. order
    C. data type
    D. number

54. Searching is a fundamental problem associated with arrays that seeks to:
    A. insert into the array the target element
    B. locate in the array where target element is
    C. reorder the elements in the array
    D. all of these are correct answers

55. What is wrong with the following code fragment?

```
...
int x=4,y=3; // line 1
const int i = 10; // line2
x = x + i; // line 3
i = y*2; // line 4
...
```

    A. Variables are not declared before use
    B. There is an error in line 3
    C. There is an error in line 4
    D. There cannot be 3 variables in such a simple program

56. Which of the following is the correct operator to compare two variables?
    A. :=
    B. =:
    C. =
    D. ==

57. Which of the following correctly declares an array?
    A. `int anarray[ 10 ];`
    B. `int anarray;`
    C. `struct array[ 10 ];`
    D. `anarray[ 10 ];`

58. Which of the following represents the correct logic for a read-until-input-failure loop?

    A.
```
while ( input succeeded )
{
// read next data item
// process last data read
}
```

    B.
```
while ( input succeeded )
{
// process last data read
// read next data item
}
```

    C.
```
// read first data item
while ( input succeeded )
{
// process last data read
// read next data item
}
```

    D.
```
// read first data item
while ( input succeeded )
{
// process last data read
}
```

59. A call-by-reference parameter is indicated by attaching the _____ to the end of the type name in a function call.
    A. Semicolon sign, ;
    B. Ampersand sign, &
    C. Hyphen sign, -
    D. None of the above

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4. Write a function called MatrixTranspose to find the transpose of multi-dimensions array. The header of the function as follow: [5 marks]

```
void MatrixTranspose (int mat[][COLUMN_SIZE], int transpMat[]
[COLUMN_SIZE], int rowSize)
```

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5. The data in the array *myList* is already sorted. Explain how Binary Search can be used to find an element in myList. [5 marks]

index:	[0]	[1]	[2]	[3]	[4]	[5]	[6]	[7]
myList	5	7	16	24	30	45	55	62

6. Write C++ statement(s) that accomplish the following:

A. Declare int variables x and y. Initialize x to 25 and y to 18 [7 marks]

B. Write a for loop which prints the integers from 100 down to 0 by fives; that is, the first three lines should be 100, 95, 90...

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C. You have an integer variable x that holds a four-digit positive integer. Write C++ code to store the middle two digits into integer variable y.

D.
```
x = 10; y = 40;
if (x >= 10) {
     if (y < 40) {
          y++; }
     } else {
          y--; }
cout << x << " " << y << endl;        //10 40
```

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Under what condition(s) would y be incremented?

E. Suppose that every time `currentTime()` is called, it returns the current time in the form of an integer (typically the number of seconds since a certain day in 1970). Using `currentTime()` and `motors()`, write C++ code to make the robot move forward at half speed for 45 seconds.

F. Rewrite the following C++ code segment to make it more efficient and to remove any unnecessary code.

```
if (x < y || y >= x) { cout << "Not good.\n"; }
cout << "Carry on.\n";
```

G. What is the output when the following code fragment is executed?

```
char ch; char title[] = "Titanic";
ch = title[1];  title[3] = ch;
cout << title << endl;
cout << ch << endl;
```

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7. Define a two-dimensional array named *temp* of three rows and four columns of type *int* such that the first row is initialized to 6, 8, 12, 9; the second row is initialized to 17, 5, 10, 6; and the third row is initialized to 14, 13, 16, 20. Answer the following questions:

   A. Write **C++ statements** for the description above. **[2 marks]**

   B. Change the contents of the last row and first column element of temp to hold the value of the second row and first column element of temp plus 7.

8. State what would be output by each of the following C++ program fragments **[3 marks]**

```
int i = 0, j = 5, a[] = { 1, 2, 3, 4, 5, 6 };while (i<4)  cout << a
[i++] << ', ' << a[--j] << endl;
```

9. Answer the following questions:

   A. Differentiate between the Enumeration (*enum*) and the Structure (*struct*) types. **[6 marks]**

   B. How is Structure (*struct*) different from Array in C++?

   C. Rewrite the following code fragment so that it uses a "do...while..." loop to accomplish the same task.

```
int n; cout << "Enter a non-negative integer: ";
cin >> n;
while (n < 0) {
 cout << "The integer you entered is negative." << endl;
 cout << "Enter a non-negative integer: ";
 cin >> n; }
```

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